

Identity Collapse: A Measurable Degradation Regime in Bearing and Battery Systems

Shawn Barnicle
Barnicle AI Systems
`ShawnBarnicle.ai@gmail.com`

May 2026

Abstract

We study adaptive threshold detection in two physical degradation settings: rolling element bearings monitored by vibration RMS and lithium-ion batteries monitored by discharge capacity. Applying a scalar identity metric derived from the ratio between baseline and current signal state, we find that the two domains occupy structurally different degradation regimes. Batteries degrade while preserving high identity ($I \approx 0.58$ – 0.72 in our sample). Bearings undergo identity collapse, losing most of their baseline signal character before failure (I as low as 0.04 in our sample). More importantly, a system can be deep in identity collapse while its temporal coherence remains high, as shown by Bearing1.1 ($I = 0.108$, $\rho = 0.978$). These are not the same quantity, and the adaptive threshold formula treats them as separate terms.

This distinction has a direct consequence for threshold correction. Linear identity modulation collapses the correction multiplier too far toward zero in the extreme-collapse regime, making it ineffective. In the documented Bearing2.4 development path, linear modulation yielded $F1 = 0.455$; square-root identity dampening preserved a stronger correction and yielded $F1 = 0.957$ on the same bearing. Across the tested XJTU-SY bearing set, all 10 validated passing bearings fall below an empirical rescue boundary near $I \approx 0.25$, while the reported failing cases lie outside the effective rescue regime or lack a usable degradation signal. All three NASA batteries pass with strong $F1$ scores (0.949 – 0.975) using a domain-specific coupling constant $\alpha = 0.034$, compared to $\alpha = 0.1$ for bearings. We document a failure case in which an incorrect α setting caused $F1$ drops of 37–43 percentage points before correction.

All experiments use real published datasets with no synthetic data. We report explicit negative results, including five bearings for which the correction did not succeed and one domain in which an incorrect α setting caused large performance drops before correction.

1 Introduction

Degradation detection methods often treat failing systems as if they degrade in structurally similar ways (Jardine et al., 2006; Lei et al., 2019). A sensor monitors a signal, compares it to a threshold, and triggers an alarm when the signal crosses that threshold. The threshold may be static or adaptive, but the underlying assumption is often the same: a degrading system is a degrading system, and the detection logic does not need to distinguish between different degradation regimes.

This paper argues that assumption is wrong in a measurable and practically important way.

We study two physical degradation settings: rolling element bearings monitored by vibration RMS (Wang et al., 2020), and lithium-ion batteries monitored by discharge capacity (Saha and Goebel, 2007). Both are well-studied systems with published benchmark datasets. Both undergo

degradation that is detectable in principle. But when we compute a scalar identity metric (instantiated as baseline RMS / current RMS for bearings and current capacity / initial capacity for batteries) the two domains tell a sharply different story. Batteries degrade while preserving high identity ($I \approx 0.71$ in our sample). Bearings undergo identity collapse, losing most of their baseline signal state before failure (I as low as 0.04 in our sample). More importantly, a system can be deep in identity collapse while its temporal coherence remains high. Bearing1_1 in the XJTU-SY dataset ends its life at $I = 0.108$ and $\rho = 0.978$ simultaneously. These are not the same quantity.

This distinction matters for adaptive threshold detection. Standard adaptive threshold formulas modulate a detection threshold using a signal-derived multiplier. When that multiplier uses identity linearly, the correction becomes ineffective in extreme identity collapse because the multiplier collapses too far toward zero. In the documented Bearing2_4 development path, linear identity modulation at $I = 0.061$ corresponds to a 94% reduction in the correction term, while square-root identity dampening reduces this to 76% and restores successful detection. In the tested XJTU-SY bearing set, all 10 validated passing bearings fell below approximately $I \approx 0.25$, while the validator documentation reported no successful rescue among the tested failing bearings at or above this regime or in no-signal cases.

This paper makes the following contributions:

- We identify identity collapse as a measurable degradation regime that is empirically distinct from identity-preserving degradation in the tested bearing and battery settings.
- We show that identity and temporal coherence are empirically distinct quantities that must be kept as separate terms in any formula that uses both.
- We show that linear identity modulation becomes ineffective in extreme collapse because the multiplier collapses too far toward zero, and that square-root dampening is the successful correction in our validated bearing set.
- We identify an empirical rescue boundary near $I \approx 0.25$ in the tested XJTU-SY bearing set: all 10 validated passing bearings fall below this range, while the reported failing cases do not show the same rescue behavior. We treat the broader intermediate- to higher-identity regime as incompletely characterized rather than resolved.
- We show that the coupling constant α must be tuned per domain, and document a failure case in which an incorrect α setting caused large performance drops before correction.

All experiments use real published datasets with no synthetic data. We report explicit negative results, including five bearings for which the correction did not succeed and one domain in which an incorrect α setting caused large performance drops before correction.

2 The Adaptive Threshold Framework

2.1 Motivation

Standard degradation detection compares a monitored signal against a fixed threshold. When a system degrades slowly or unevenly, fixed thresholds produce either too many false alarms early in the degradation process or too many missed detections late in it. Adaptive threshold methods address this by modulating the threshold as a function of observed signal behavior. The key design question is: what signal properties should drive that modulation, and how should they be combined?

The framework studied here uses two properties: identity (how much the system’s current signal state resembles its baseline) and temporal coherence (how smoothly the signal is evolving over time). These two properties are distinct, and the formula we validate treats them as separate terms. We study this framework on 10 mechanical bearing systems from the XJTU-SY dataset and 3 electrochemical battery systems from the NASA battery dataset.

2.2 Identity

We define identity I as the ratio of a system’s baseline signal state to its current signal state, normalized so that $I = 1$ at baseline and I decreases as the system degrades. The domain-specific instantiations are:

$$I_{\text{bearing}} = \frac{\text{baseline RMS}}{\text{current RMS}} \quad (1)$$

$$I_{\text{battery}} = \frac{\text{current capacity}}{\text{initial capacity}} \quad (2)$$

For bearings, RMS increases as vibration amplitude grows during degradation, so identity falls. For batteries, capacity decreases monotonically, so identity also falls. In both cases $I \in (0, 1]$ during normal degradation, with lower values indicating greater departure from baseline behavior.

2.3 Temporal Coherence

We define temporal coherence ρ as the Pearson autocorrelation of the monitored signal time series at lag 1:

$$\rho = \text{corr}(\text{signal}[t], \text{signal}[t - 1]) \quad (3)$$

Higher ρ indicates that the signal is evolving smoothly from one measurement to the next. Lower ρ indicates erratic or discontinuous evolution. As we show in Section 4, I and ρ are empirically distinct: a system can have very low identity and still have high temporal coherence, as in the case of Bearing1_1 ($I = 0.108$, $\rho = 0.978$).

2.4 The Adaptive Threshold Formula

The adaptive threshold at any point in a system’s lifecycle is:

$$\text{Threshold} = 1 + \alpha \times \left(\frac{\text{degradation rate}}{100} \right) \times \sqrt{I} \times \rho \quad (4)$$

where α is a domain-specific coupling constant (bearings: $\alpha = 0.1$, batteries: $\alpha = 0.034$), degradation rate is expressed on a percentage scale, and $\sqrt{I}$ is the square root identity dampening term.

The formula has three components that interact:

- $\sqrt{I}$ scales the correction in proportion to identity loss, but with non-linear dampening that prevents the multiplier from collapsing to near zero during extreme identity collapse.
- ρ scales the correction by temporal smoothness, reducing threshold adjustment when the signal is evolving erratically.
- α sets the overall correction magnitude for the domain.

2.5 Why Square Root and Not Linear

The choice of $\sqrt{I}$ rather than I as the identity term is the central mathematical decision in this framework. With a linear term, the multiplier $I \times \rho$ collapses toward zero as identity collapses, making the correction increasingly ineffective in the extreme-collapse regime. With a square root term, the multiplier $\sqrt{I} \times \rho$ falls more slowly, maintaining a meaningful correction level even at very low identity values.

The concrete comparison at $I = 0.061$, $\rho = 0.988$ (Bearing2_4):

- Linear multiplier: $0.061 \times 0.988 = 0.060 \rightarrow 94\%$ reduction in the correction term
- Square root multiplier: $\sqrt{0.061} \times 0.988 = 0.244 \rightarrow 76\%$ reduction in the correction term.

The linear version drives the multiplier so low that the correction becomes ineffective in the extreme-collapse regime, leading to poor detection performance ($F1 = 0.455$). In the documented Bearing2_4 development path, square-root dampening preserved a stronger correction and produced the successful rescue result ($F1 = 0.957$) on the same bearing.

2.6 What This Framework Measures and What It Does Not

The adaptive threshold formula measures whether a system’s current signal state has departed from its baseline, and how smoothly that departure is evolving. It does not measure whether a system is in a specific failure mode, nor does it classify the type of degradation occurring. It is a detection instrument, not a diagnostic one.

In principle, a system that has shifted to a new stable operating regime could produce a low identity value without being on a failure trajectory, and the framework does not resolve that ambiguity. We discuss this boundary condition further in Section 10.

3 Identity Collapse as a Distinct Degradation Regime

3.1 Two Ways a System Can Degrade

Not all degrading systems lose their baseline signal character at the same rate or in the same way. In our sample, the two domains we studied tell a sharply different story when measured by identity.

Battery B0005 from the NASA dataset shows high end-state identity ($I = 0.714$) and very high temporal coherence ($\rho = 0.998$), indicating smooth identity-preserving degradation. The system loses capacity over its monitored lifetime, but its fundamental signal character remains largely intact.

Bearing1_1 from the XJTU-SY dataset tells a different story. Its RMS vibration amplitude increases by a factor of 9.26 over its monitored lifetime, driving its end-state identity down to $I = 0.108$. The system has undergone a substantial loss of baseline signal identity before failure. Yet its temporal coherence remains high at $\rho = 0.978$. This is identity collapse: a substantial loss of baseline signal character that proceeds smoothly over time.

3.2 Why the Distinction Matters

The difference between these two regimes is not just descriptive. It has a direct consequence for adaptive threshold detection.

In the identity-preserving regime, the signal magnitude changes gradually and the system’s behavior remains recognizable relative to baseline. Standard threshold methods and mild adaptive

corrections both work reasonably well in this regime. The three NASA batteries in our validated set all fall here, with end-state identity values of 0.583, 0.714, and 0.723, and all three pass with strong $F1$ scores (0.967, 0.975, and 0.949 respectively) even with a small coupling constant of $\alpha = 0.034$.

In the identity-collapse regime, the signal has transformed so far from its baseline that a standard threshold no longer captures the degradation structure. The correction required is larger, and its scaling must account for the depth of the collapse. As we show in Section 5, this is exactly where linear identity modulation fails and square-root dampening succeeds.

3.3 Identity as a Regime Signal

A practical consequence of this finding is that identity I , computed directly from baseline and current signal measurements, may serve as a regime indicator before applying any correction. In our bearing set, the empirical boundary near $I \approx 0.25$ separates the cases where the square-root correction succeeded from the cases where it did not. We return to this point in Section 6.

We note that this boundary is observed in the tested XJTU-SY bearing set and is not claimed as a universal threshold across domains or system types. The battery domain operates in a structurally different regime and does not appear to require rescue-style correction in the validated battery set.

4 Identity and Temporal Coherence are Distinct

4.1 The Key Observation

A natural question when proposing two separate signal properties is whether they are actually measuring different things. If identity and temporal coherence were tightly correlated, keeping them as separate terms in the formula would add complexity without adding information.

The bearing and battery examples show that they are not the same quantity.

System	Identity I	Temporal Coherence ρ
Battery B0005	0.714	0.998
Bearing1_1	0.108	0.978

Table 1: End-state identity and temporal coherence for two representative systems. Both systems have high ρ ; only the bearing has low I .

Both systems have high temporal coherence. Only the bearing has low identity. A single composite signal combining the two would assign similar values to both systems at the coherence dimension while masking the sharp difference at the identity dimension. That masked difference is precisely the information needed to decide whether and how strongly to apply the adaptive correction.

4.2 What Each Term Captures

Identity I captures the magnitude relationship between current and baseline signal state. It answers the question: how far has this system traveled from where it started?

Temporal coherence ρ captures the smoothness of the signal’s evolution from one measurement to the next. It answers the question: is this system changing erratically or steadily?

A system can answer these two questions very differently. Bearing1_1 has traveled very far from its baseline ($I = 0.108$) but is doing so in a smooth and continuous way ($\rho = 0.978$). Treating these as the same quantity would produce a misleading summary of its state.

4.3 Scope of This Observation

This conclusion is illustrated clearly by the two diagnostic examples above and is compatible with the broader bearing–battery contrast seen in the validated sets. It was not established through a dedicated independence test across the full dataset, and broader testing of the relationship between I and ρ across system types remains open. We include it here because it provides the empirical justification for keeping the two terms separate in the adaptive threshold formula.

5 Square-Root Dampening in the Extreme-Collapse Regime

5.1 The Problem with Linear Identity Modulation

The intuitive design choice for an identity-aware adaptive threshold is to scale the correction linearly with identity loss. As identity falls, the multiplier falls proportionally, and the threshold adjusts accordingly. This can behave reasonably in less severe degradation, where identity has not yet collapsed and the multiplier remains in a useful range.

In the extreme-collapse regime, however, linear scaling breaks down. When identity drops to very low values, as it does in several of the XJTU-SY bearings in our sample, the linear multiplier $I \times \rho$ collapses toward zero. The correction becomes too weak to adapt meaningfully to the actual degradation state of the system, and the adaptive threshold loses much of the benefit of the identity-aware adjustment.

The concrete example is Bearing2_4, which enters end-of-life with $I = 0.061$ and $\rho = 0.988$:

- Linear multiplier: $0.061 \times 0.988 = 0.060$
- This corresponds to a 94% reduction in the correction term.
- In the documented Bearing2_4 development path, the earlier linear formulation yielded $F1 = 0.455$.

The linear version does not preserve a useful correction in the extreme-collapse regime. The multiplier collapses so far toward zero that the adaptive threshold loses much of the benefit of the identity-aware adjustment.

5.2 Square-Root Dampening as the Correction

The solution adopted in this framework is to replace the linear identity term with its square root. The square root function maps very low identity values to higher multiplier values than the linear version, preserving a meaningful correction even in the extreme-collapse regime.

For the same Bearing2_4 example:

- Square-root multiplier: $\sqrt{0.061} \times 0.988 = 0.247 \times 0.988 = 0.244$
- This corresponds to a 76% reduction in the correction term.
- In the documented Bearing2_4 development path, the square-root formulation yielded $F1 = 0.957$.

The difference between a 94% reduction and a 76% reduction may appear small in absolute terms, but it is the difference between a multiplier of 0.060 and a multiplier of 0.244, a fourfold difference in the effective correction applied to the threshold. In the documented Bearing2_4 development path, this difference was the deciding factor between detection failure and a strong rescue result.

5.3 Why Square Root and Not Other Non-Linear Functions

The square root was identified empirically as the non-linear choice that produced a workable operating point in the validated bearing set. Two alternatives were explored during development and rejected:

- **Linear** $I \times \rho$: the multiplier collapses too far toward zero in extreme collapse, as described above.
- **Identity velocity** dI/dt : explored as a supplementary term but provided no additional benefit over $\sqrt{I}$ and was not incorporated into the final formula.

We do not claim that $\sqrt{I}$ is the uniquely correct non-linear function for all possible systems or domains. We report that it was the function that worked in the validated bearing set and that produced sensible behavior in the battery set with a domain-specific α .

5.4 Validation Results Under Square-Root Dampening

Table 2 summarizes the identity values and $F1$ scores for all 10 validated passing bearings under the square-root formula. All 10 fall below $I \approx 0.25$ and all 10 pass the $F1 \geq 0.5$ detection threshold.

Test	System	Identity I	$F1$ Score
1	Bearing1_1	0.108	0.756
2	Bearing1_2	0.099	0.780
3	Bearing1_3	0.083	0.874
4	Bearing1_4	0.248	0.550
5	Bearing2_2	0.084	0.550
6	Bearing2_3	0.110	0.610
7	Bearing2_4	0.061	0.957
8	Bearing2_5	0.087	0.570
9	Bearing3_1	0.044	0.852
10	Bearing3_4	0.118	0.920

Table 2: Identity values and $F1$ scores for all 10 validated passing bearings under square-root dampening. All 10 fall below $I \approx 0.25$. Pass criterion: $F1 \geq 0.5$.

6 The Empirical Rescue Boundary Near $I \approx 0.25$

6.1 An Observed Pattern in the Tested Set

Across the 15 XJTU-SY bearings in our tested set, the clearest observed pattern is that all 10 validated passing bearings fall below approximately $I \approx 0.25$. The reported failing cases either lie outside the effective rescue regime or lack a usable degradation signal.

6.2 What the Boundary Represents

The $I \approx 0.25$ boundary is not a hard cutoff derived from first principles. It is an empirically observed separation point in the tested bearing set. Below this value, the system has lost enough baseline signal identity that the square-root correction provides a meaningful and beneficial adjustment to

the threshold. Above this value, the additional square-root correction was not observed to show the same rescue benefit in the tested set.

This has a practical consequence: identity I , computed directly from baseline and current signal measurements, can serve as a preliminary indicator of whether the rescue-style adjustment is likely to help. A system entering the monitoring pipeline with $I < 0.25$ is a candidate for the square-root correction. In the tested bearing set, systems at or above this regime were not observed to benefit from rescue-style correction.

6.3 Scope and Limitations of This Boundary

We state clearly that this boundary is observed in the tested XJTU-SY bearing set and is not claimed as a universal threshold. Several caveats apply:

- The behavior of the intermediate- to higher-identity bearing regime remains incompletely characterized. The validated passing bearings all fall below approximately $I \approx 0.25$, but the broader failing set does not by itself establish a sharp mechanistic transition across all bearings.
- The battery domain operates at much higher identity values (0.583 to 0.723) and is not subject to this boundary. Batteries pass with strong $F1$ scores using a small $\alpha = 0.034$ without rescue-style correction.
- Whether this boundary holds for other mechanical systems, other operating conditions, or other sensor modalities is an open question.

The filed provisional patent states that identity values below approximately 0.25 indicate systems for which the method is effective, which is consistent with the pattern observed here. We present this as an empirical finding from the validated set rather than a derived theoretical result.

7 Domain-Specific Alpha

7.1 Alpha as a Domain Parameter

The coupling constant α in the adaptive threshold formula controls the overall magnitude of the identity-aware correction. A larger α produces a stronger threshold adjustment for a given identity and coherence value; a smaller α produces a subtler one. The formula structure does not prescribe a single value of α across all domains, and the validated results in this paper use two distinct values: $\alpha = 0.1$ for bearings and $\alpha = 0.034$ for batteries.

This is not only a tuning detail. In the validated domains, it appears to reflect a physical difference between the two degradation mechanisms. Bearing vibration RMS can grow dramatically during failure, whereas battery capacity decays gradually and monotonically. A correction of the same magnitude as the bearing correction would over-adjust in the battery domain, where the system retains high identity throughout its monitored lifetime.

7.2 The Alpha Failure Case

The domain-specificity of α is not a hypothetical concern. Early battery tests in the development of this framework used $\alpha = 1.0$, the same order of magnitude as was used in initial bearing experiments. The result was a drop in $F1$ of 37 to 43 percentage points across the tested batteries before the error was identified and corrected.

This failure was caused entirely by the wrong α , not by a flaw in the formula structure. Once α was corrected to 0.034, all three validated batteries maintained or improved their $F1$ scores relative to the baseline formula. The formula itself was not broken; the domain calibration was.

We report this failure case explicitly because it has a practical implication: deploying this framework to a new domain requires establishing the correct α through validation on that domain before drawing conclusions about the method’s effectiveness. An incorrect α can make a working method appear to fail.

7.3 What Alpha Encodes

We interpret α as encoding the physical character of degradation in each domain, specifically how strongly the identity-aware correction should respond to a given level of identity loss. In a domain where identity collapses dramatically and the signal transforms fundamentally, a larger α is needed to drive a threshold adjustment that keeps pace with the degradation. In a domain where identity is largely preserved and the signal evolves smoothly, a smaller α keeps the correction proportionate to the smaller adjustment needed.

This interpretation is consistent with both validated domains. It does not prescribe how α should be set for a new domain; that requires empirical validation. We do not claim that $\alpha = 0.1$ is correct for all mechanical systems or that $\alpha = 0.034$ is correct for all electrochemical systems. We report the values that worked in the tested sets.

8 Negative Results

8.1 Five Bearings for Which the Correction Did Not Succeed

Of the 15 XJTU-SY bearings in our tested set, five did not pass the $F1 \geq 0.5$ detection threshold under square-root dampening. We report these failures explicitly because they define the boundary conditions of the method and make the positive results more credible.

The five failing bearings fall into two categories:

- **No usable degradation signal.** One bearing showed no meaningful degradation structure in the data. The formula cannot create detection signal where none exists in the underlying measurements. We interpret this as a boundary condition of the method rather than as evidence against the formula itself.
- **Outside the effective rescue regime.** The remaining failing bearings lay outside the effective rescue regime described in the validator documentation, including high-identity cases and at least one case in which the correction hurt performance. This confirms that the correction is not neutral when applied outside the effective regime: it can cause harm.

These results are not hidden in the repository. The validator documentation explicitly lists failing configurations in a separate directory and describes the reason for each failure. We consider this transparency part of the scientific record.

8.2 The Wrong Alpha on Batteries

As described in Section 7, early battery tests using $\alpha = 1.0$ produced $F1$ drops of 37 to 43 percentage points. This is a documented failure in the development path. We include it here as a negative result rather than a footnote because it carries scientific information: the method is sensitive to

α calibration, and an uncalibrated deployment should be expected to produce results like these rather than the strong results reported for the validated battery set.

8.3 The Moderate-Identity Regime Remains Unclear

The current bearing results do not cleanly resolve behavior in the intermediate-identity range. The strongest observed pattern is that all 10 validated passing bearings fall below approximately $I \approx 0.25$, while the reported failing cases lie outside the successful rescue set or lack a usable degradation signal. We therefore do not make a stronger claim about how the square-root correction behaves throughout the broader moderate-identity regime.

8.4 What These Negative Results Show

Taken together, the negative results in this paper establish three boundary conditions for the method:

- The square-root correction requires a usable degradation signal in the data.
- The square-root correction is most effective in the low-identity regime observed in the tested bearing set, and is not observed to show the same rescue benefit outside that regime.
- The coupling constant α must be calibrated per domain before deployment.

None of these boundary conditions undermine the positive findings. They define the scope within which those findings hold.

9 Cross-Domain Validation: Batteries as the Contrast Case

9.1 Setup

To test whether the adaptive threshold framework transfers beyond the bearing domain, we applied it to three lithium-ion battery systems from the NASA battery dataset: B0005, B0006, and B0018. Each battery was monitored by discharge capacity over repeated charge-discharge cycles until end of life. Identity was computed as current capacity divided by initial capacity, and temporal coherence was computed as the Pearson autocorrelation of the capacity time series at lag 1.

The same formula structure was used as for bearings, with one change: $\alpha = 0.034$ rather than $\alpha = 0.1$. As discussed in Section 7, this smaller value reflects the more gradual and identity-preserving character of battery degradation.

9.2 Results

Table 3 summarizes the identity, temporal coherence, and $F1$ scores for the three validated batteries.

All three batteries pass with strong $F1$ scores. Unlike the bearing results, where the square-root correction provided the decisive rescue in the low-identity regime, the battery results show a different pattern: high identity is preserved throughout, temporal coherence is very high, and the correction provides a small positive adjustment rather than a large rescue. The effect of the formula in this domain is stabilizing rather than rescuing.

System	Identity I	Coherence ρ	$F1$ Score
B0005	0.714	0.998	0.967
B0006	0.583	0.996	0.975
B0018	0.723	0.989	0.949

Table 3: Identity, temporal coherence, and $F1$ scores for three NASA battery systems under the adaptive threshold formula with $\alpha = 0.034$. Pass criterion: $F1 \geq 0.5$.

9.3 What the Battery Results Add

The battery results serve two functions in this paper. First, they show that the formula structure transfers to a second physical domain with appropriate α calibration. Second, and more importantly, they provide the contrast case that makes the bearing story legible.

Without the battery results, the bearing findings would describe only one kind of system. With the battery results, we can see that identity-preserving degradation and identity-collapse degradation are genuinely different regimes that produce different detection behaviors under the same formula. Batteries degrade smoothly, retain their signal character, and pass without rescue-style correction. Bearings lose their signal character dramatically and require the square-root correction to pass. The formula produces different observed behaviors in the two domains.

10 Discussion, Limitations, and Future Work

10.1 What This Work Shows

The central finding of this paper is that not all degrading physical systems fail the same way, and that this difference is measurable and practically important. In our tested sets, bearings undergo identity collapse, a substantial loss of baseline signal character that proceeds smoothly over time, while batteries degrade while largely preserving their signal identity. These two regimes require different threshold correction strategies, and the same formula structure can serve both only if α is calibrated per domain.

The square-root dampening result is the sharpest methodological finding. Linear identity modulation collapses the correction multiplier too far toward zero in the extreme-collapse regime, making it ineffective. Square-root dampening preserves a meaningful correction at low identity values and restores detection performance in the tested bearing set. In the documented Bearing2_4 development path, this difference was the deciding factor between $F1 = 0.455$ and $F1 = 0.957$.

The empirical rescue boundary near $I \approx 0.25$ in the tested bearing set has a practical consequence: identity, computed directly from signal measurements before any correction is applied, can serve as a regime indicator. In the tested set, systems below this boundary were candidates for the square-root rescue correction; systems above it were not observed to benefit.

10.2 Limitations

Sample size and scope. The bearing results are based on 15 systems from one dataset under three operating conditions. The battery results are based on three systems. These are small samples, and the patterns we report, including the $I \approx 0.25$ boundary, may not hold across a broader range of systems, operating conditions, or sensor modalities.

The intermediate- to higher-identity bearing regime is incompletely characterized. The validated passing bearings all fall below approximately $I \approx 0.25$, but the broader failing set

does not by itself establish a sharp mechanistic transition across all higher-identity bearings. We therefore treat the behavior of the intermediate- to higher-identity bearing regime as incompletely characterized rather than resolved.

Alpha calibration is required. The framework does not prescribe α for a new domain. Deploying it without domain-specific validation risks the kind of performance collapse we documented in the early battery experiments. This is a practical limitation for any application to a new system type.

Single sensor modality per domain. Bearings were monitored by vibration RMS and batteries by discharge capacity. Whether the same formula and regime logic apply to other sensor modalities such as temperature, acoustic emission, or impedance remains untested.

The identity-coherence independence observation is illustrative, not formally established. The observation that I and ρ are distinct is supported clearly by the bearing and battery diagnostic examples, but was not established through a dedicated independence test across the full validated set.

10.3 Future Work

The most direct extension of this work is to test the framework on additional mechanical and electrochemical systems to determine whether the $I \approx 0.25$ boundary holds beyond the XJTU-SY bearing set. A larger and more diverse bearing dataset would also allow the intermediate- to higher-identity regime to be better characterized.

A second direction is to test the framework on other physical degradation domains such as turbofan engines, electrical grids, or solar panels, where run-to-failure data exists and identity collapse may occur at different rates or scales.

A third direction is to investigate whether α can be estimated from early-lifecycle signal properties rather than requiring full validation runs. If α could be set automatically from the character of the degradation trajectory, the framework would be more practical to deploy in new domains.

Finally, the observation that identity I may serve as a regime indicator before correction is applied raises the question of whether a two-stage detection architecture, first assessing regime and then applying the appropriate correction, would outperform the single-formula approach validated here. We leave that as an open question.

References

- Wang, B., Lei, Y., Li, N., and Li, N. A hybrid prognostics approach for estimating remaining useful life of rolling element bearings. *IEEE Transactions on Reliability*, 69(1):401–412, 2020.
- Saha, B. and Goebel, K. Battery data set. NASA Ames Prognostics Data Repository, NASA Ames Research Center, Moffett Field, CA, 2007. <https://www.nasa.gov/content/prognostics-center-of-excellence-data-set-repository>
- Lei, Y., Li, N., Guo, L., Li, N., Yan, T., and Lin, J. Machinery health prognostics: A systematic review from data acquisition to RUL prediction. *Mechanical Systems and Signal Processing*, 104:799–834, 2019.
- Jardine, A., Lin, D., and Banjevic, D. A review on machinery diagnostics and prognostics implementing condition-based maintenance. *Mechanical Systems and Signal Processing*, 20(7):1483–1510, 2006.

Patent Disclosures

The methods described in this paper are the subject of U.S. Provisional Patent Application No. 63/921,348 (filed November 20, 2025). No license to implement or commercialize the described methods is granted by this publication. All rights reserved.